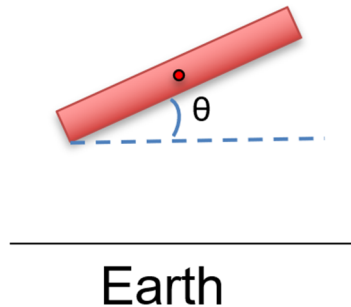
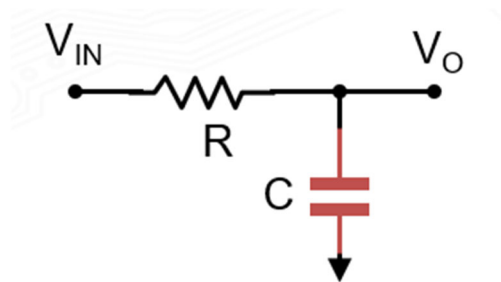


1. An accelerometer is tilted along its Y-axis as shown in the diagram.



- (a) Which axis/axes of the accelerometer will detect acting force?
 - (b) Calculate the angle Θ in terms of the accelerometer output's a_x , a_y and a_z .
- (a) Derive the expression of the low pass filter shown below that can be implemented digitally using an algorithm.



- (b) Find the coefficients value of this digital filter if
 - (i) Bandwidth of the filter = 10 Hz
Time interval in between samples computation (i.e. $[n]$ and $[n-1]$) $T = 1\text{msec}$
 - (ii) Bandwidth of the filter = 100 Hz
Time interval in between samples $T = 1\text{msec}$
3. Describe and contrast the characteristics of Accelerometer and Gyroscope.